

DATE:19th of September, 2026

SMART HOSPITAL & RESOURCE ALLOCATION SYSTEM

Project report

CSC 1012 – Introduction to Computer Programming

University of Sri Jayewardenepura

Faculty of Applied Science



Name: H.M.S.P. HERATH

Index: AS20250484

TABLE OF CONTENT

	PAGE
01.GitHub Repository.....	03
02. Project Overview.....	04
03. System Architecture.....	05
04. Data Structures.....	06
05. Development Assumptions.....	07
06. List of Implemented Functions.....	07
07. Conclusion.....	09
08. References.....	09

01.GitHub REPOSITORY

https://github.com/subhash18/hospital_system.git

02. PROJECT OVERVIEW

Objective

The main objective of this project is to create a Smart Hospital Resource Allocation System using the C programming language. The system will use a basic menu to help hospital workers with patients, beds, emergency situations, and medical expenses. It will use arrays to store patient information, functions for various tasks, and strings to handle names and other text. The system will also implement loops and conditions so that it can select and repeat tasks when needed. It will compute patient waiting time, hospital charges, discounts, and other expenses. It will also be used to control hospital beds and wards and save important information using files. Overall, this project is useful in demonstrating how basic C programming concepts can be used to build a simple hospital management system.

Key features

- Fixed tables will be used to store information on doctor specialties and hospital wards.
- Patients will be registered with automatic IDs like PAT-1001, PAT-1002, etc.
- The system will also keep a record of hospital beds in a 2D array, with 0 means available and 1 meaning occupied.
- The system will calculate the patient bills automatically, including emergency charges, ward costs, and age-based discounts.
- The system estimates the waiting time according to the number of patients waiting for the same specialty.
- Patients will be sorted by emergency level, such as critical, urgent, and normal.
- If patients have the same emergency level, they will be arranged according to their registration order.
- The system will save bed information and patient billing records in files so the information can be used later.

Technology stack

Component	Details
Programming Language	C
IDE	Code:: Blocks
Data Structure Approach	Parallel Arrays
Files	Standard C files
platform	Windows

03. SYSTEM ARCHITECTURE

Modular Design

The system is divided into different parts, and each part handles a specific task. Each part is kept in a separate .c file, while the common information is stored in a shared header file called “hospital.h”. This header file contains the main constants, arrays, and function declarations used by the program. This structure makes the program easier to understand, manage, and update. It also helps to find and fix errors.

File Breakdown

Source File	Responsibility
hospital.h	Header file, constants, extern declarations, function prototypes
main.c	Program entry point, main menu loop, start and exit
Hospital_data.c	Specialties, wards, bed occupancy matrix
Patient.c	Patient registration, input validations, bill display, patient list
billing.c	All billing and waiting time calculations
priority.c	Priority display
report.c	Performance analytics and ward occupancy percentage
File_handling.c	Saved bed status and patient records

04. DATA STRUCTURE

Patient Parallel Array

Array	Type	purpose
patientId[]	Char[20]	Generate ID (PAT-1001)
patientName[]	Char[100]	Patient Full Name
patientAge[]	int	Age in years (0-120)
urgencyLevel[]	int	1=Normal, 2=Urgent, 3=Critical
specialtyId[]	int	Selected speciality(1-4)
admitted[]	int	1=ward, 0=OPD
wardId[]	int	Ward number if admitted
daysAdmitted[]	int	Length of admitted (days)
assignedBed[]	int	Bed index
registrationOrder[]	int	Order of registration
baseFee[] to finalPayable[]	double	Billing components for each patient
waitingTime[]	double	Estimated wait in minutes

Bed Occupancy Matrix

- The system uses a 2D array called **bedOccupancy[NUM_WARDS][MAX_BEDS]** to keep track of the hospital beds.
- There are 4 wards, and each ward has up to 20 beds.
- The value **0** means the bed is free, while **1** means the bed is being used.

05. DEVELOPMENT ASSUMPTIONS

- The system can handle up to 100 patients at a time.
- Each patient gets an ID automatically, such as PAT-1001, PAT-1002, based on the order they are registered.
- Each specialty has a daily limit.
- A patient can be admitted to any ward
- Patients under 5 years old or over 65 years old receive a 15% discount on the gross total.
- The system does not have a discharge option
- The system checks the user's input and does not allow invalid inputs.

06. LIST OF IMPLEMENTED FUNCTIONS

Function	Module	Purpose
clearInputBuffer	patient.c	Remove any extra characters in the inputs so that the next input can be entered correctly
readLine	patient.c	Safe input via fgets
getIntInRange	patient.c	Validated integer input
generatePatientID	patient.c	Create identifier
registerPatient	patient.c	Complete registration
displayPatientBill	patient.c	Print admission and bill
displayAllPatients	patient.c	Shows all patient details
displaySpecialties	hospital_data.c	Shows specialty lookup table
displayWards	hospital_data.c	Shows the ward lookup table
findAvailableBed	hospital_data.c	Return first free bed index or -1

displayBedOccupancy	hospital_data.c	Print full bed status map
calculateWaitingTime	billing.c	Queue length $\times$ consultation time
calculateEmergencySurcharge	billing.c	0 / 20 % / 50 % of base fee
calculateWardCost	billing.c	Days $\times$ daily rate
calculateGrossTotal	billing.c	Sum of fee + surcharge + ward cost
calculateAgeDiscount	billing.c	15 % of gross if age <5 or > 65
calculateFinalPayable	billing.c	Gross – discount
displayPatientsByPriorit	priority.c	Bubble-sort triage ranking
displayAnalytics	report.c	Performance report
calculateWardOccupancy	report.c	Occupied % for a given ward
saveBeds / loadBeds	file_handling.c	Restore bed matrix
savePatientRecord	file_handling.c	Billing loc entry
displayMenu / main	main.c	Menu/program control flow

07. CONCLUSION

The smart hospital and resource Allocation system meets the main requirements of the CSC 1012 Assignment. It uses arrays, functions, calculations, and file handling to manage patients, beds, and billing in a simple and practical way.

08. REFERENCES

-University of Sri Jayewardenepura – Faculty of Applied Sciences, CSC 1012 Introduction to Computer Programming Assignment (Individual) – Smart Hospital & Resource Allocation System, 2026.

-CSC 1012 Lecture materials

-ChatGPT

-Code::Blocks IDE – <https://www.codeblocks.org/>

-END-